

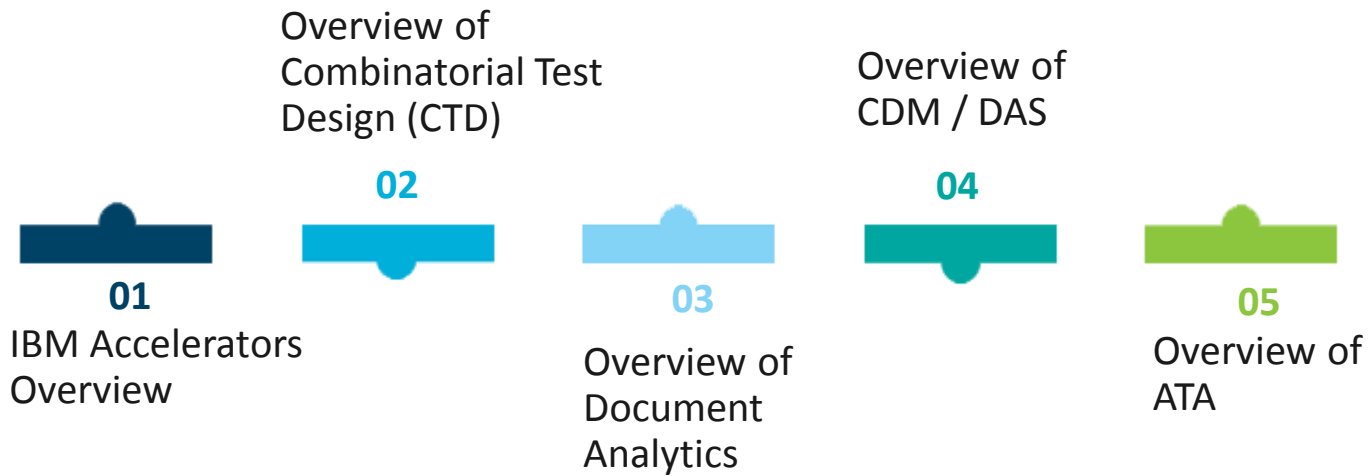


Core Testing



Day 12: Accelerators and ATA Basics





01 IBM Accelerators Overview

A c t i v i t y

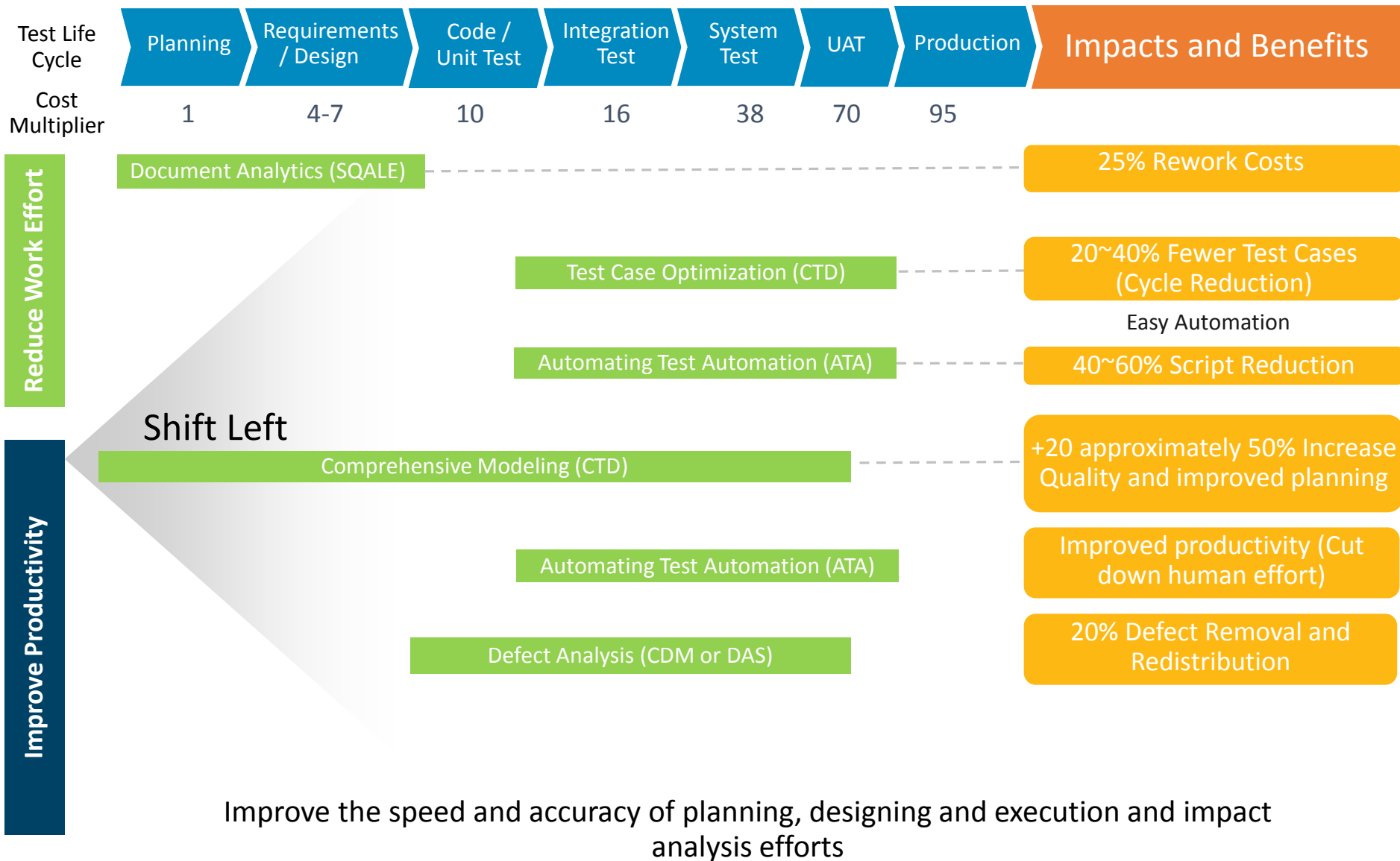
What are IBM Accelerators: Discussion

What has been the traditional mode of testing?

Why use Accelerators

What are the Next Generation Accelerators?

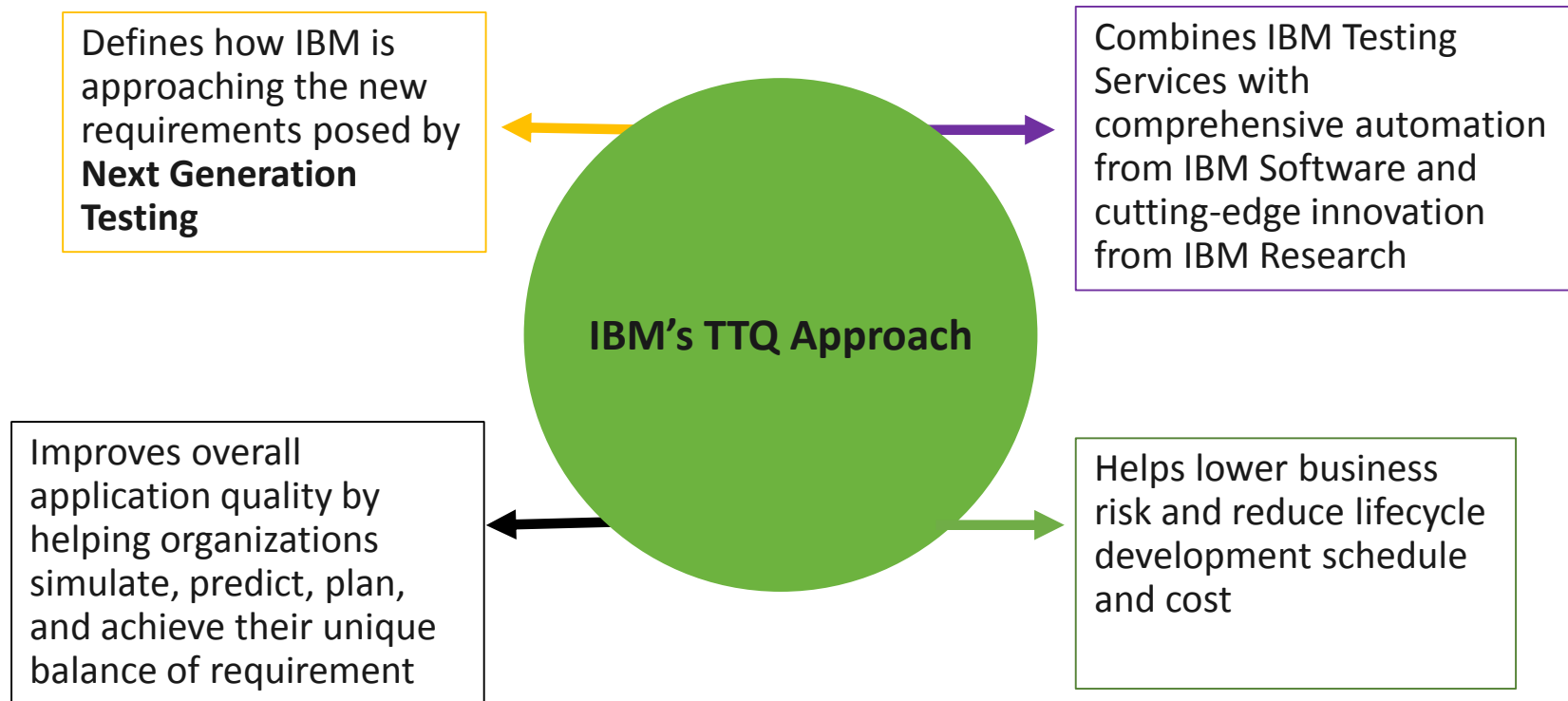
Impact of test accelerators on test lifecycle



Introduction to IBM Total Test Quality (TTQ) Accelerators



IBM's Total Test Quality approach balances **four project factors equally** to achieve most effective result from every testing penny spent, such as **Cost Reduction, Quality Improvement, Risk Management, and Schedule Acceleration**



Traditional test approach versus next generation testing



Next Generation Testing

Less Cost

Fewer test scripts to
execute \ automate

Good Quality

Less test effort

Traditional test approach

Higher cost

More test scripts to
execute \ automate

Poor Quality

More test effort



IBM Total Test Quality (TTQ) Accelerators: Next Generation Testing



Advanced
Planning and
Enhanced
Design

Comprehensive Modeling solution:

- Minimizes total number of functional cases and increases quality—CTD is a primary concept.

Collaborative Defect Management (CDM):

- Statistical, objective defect classification, and analysis process
- Find and fix defects as early as possible and prevent future defects from being injected.



Predictive
Analytics



Static Test
Coverage and
Effectiveness

Document Analytics:

- Establish **model of requirements** and **automate** traceability.
- Find defects and omissions early in the lifecycle.



Questions?

01

Which component minimizes the total number of functional cases and increases quality?

A

Modeling solution

B

Collaborative Defect Management (CDM)

C

Document Analytics

D

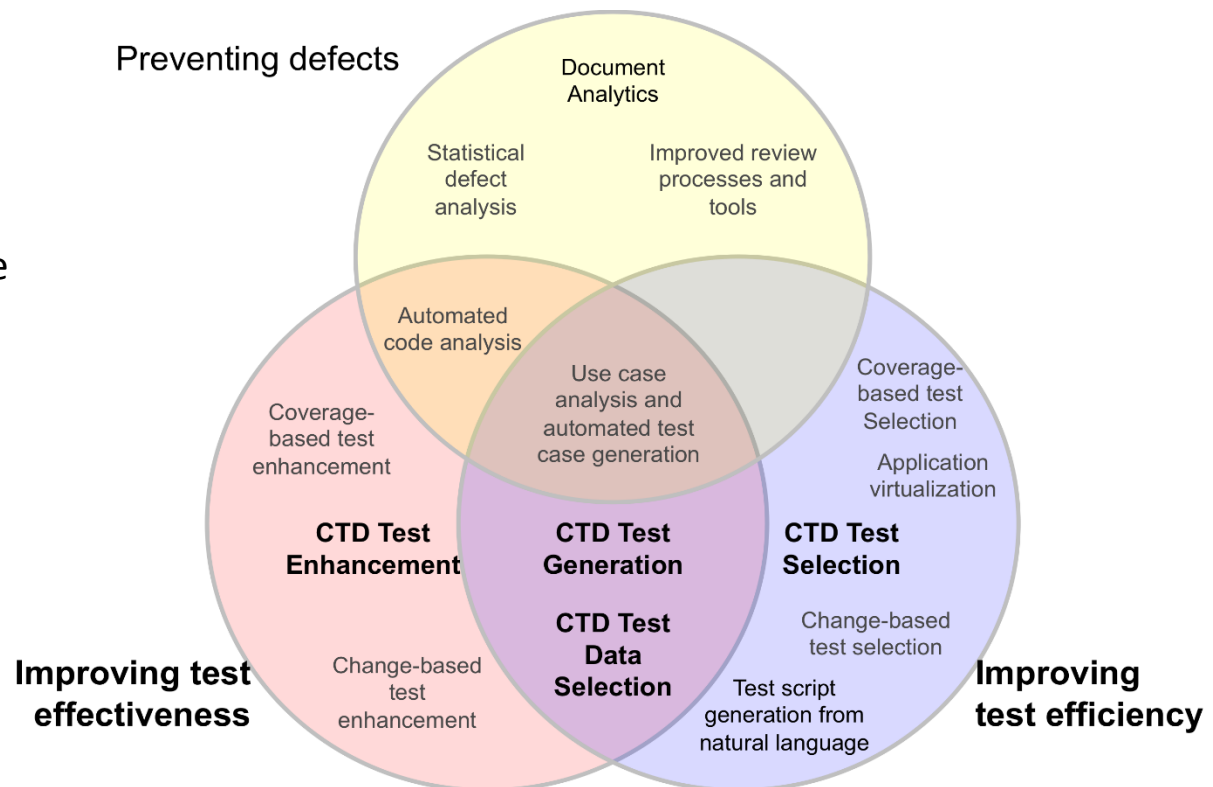
None of the above

02 Overview of CTD

Combinatorial Test Design or CTD is a concept used in model-based testing approach and to design or create test. CTD is an effective test design concept which enables testers to create abstracted, optimized test suite and ensures the coverage.

Benefits of CTD:

- Accelerates testing life cycle
- Generates effective test suites
- Improves efficiency of test suite
- Cost, time and effort reduction
- Improved time to market



What is CTD?: An illustration

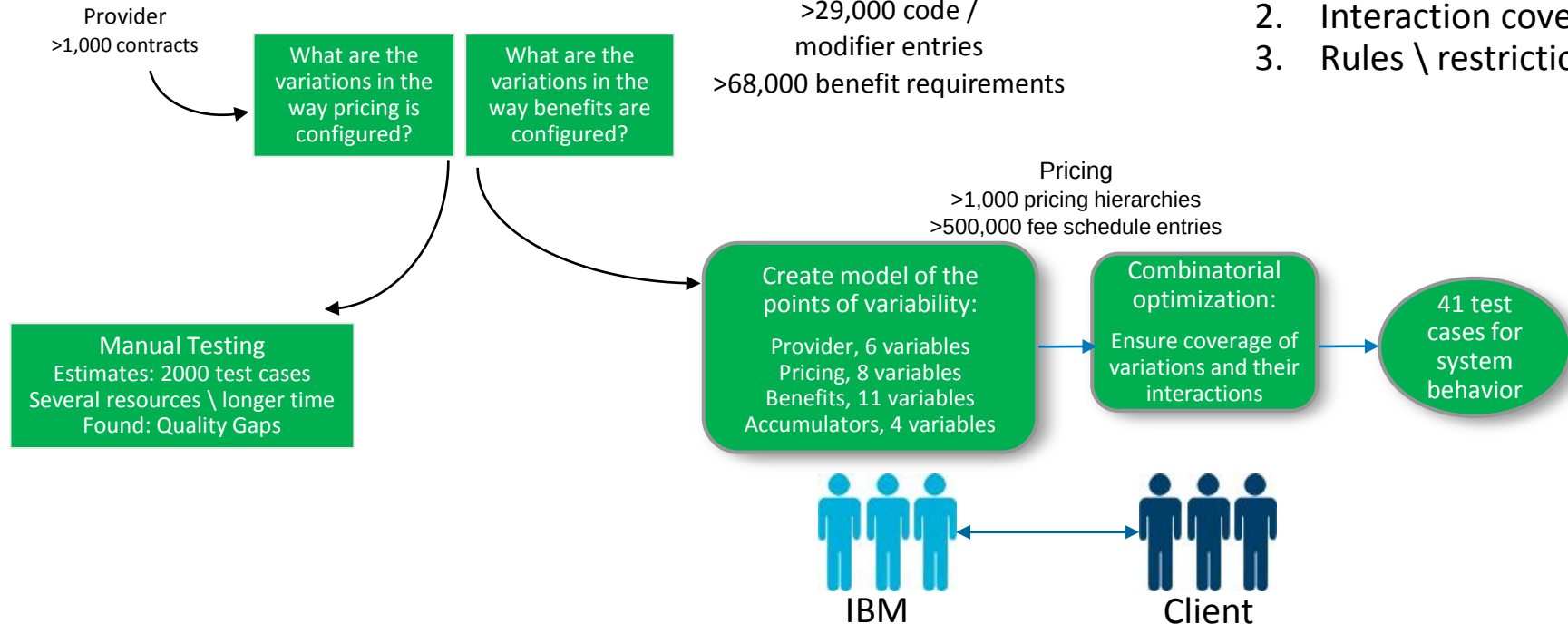
Claims Settlement

More than a trillion potential combinations

Application property
Approximately 17, 000 ICD-9codes
>29,000 code / modifier entries
>68,000 benefit requirements

Important aspects:

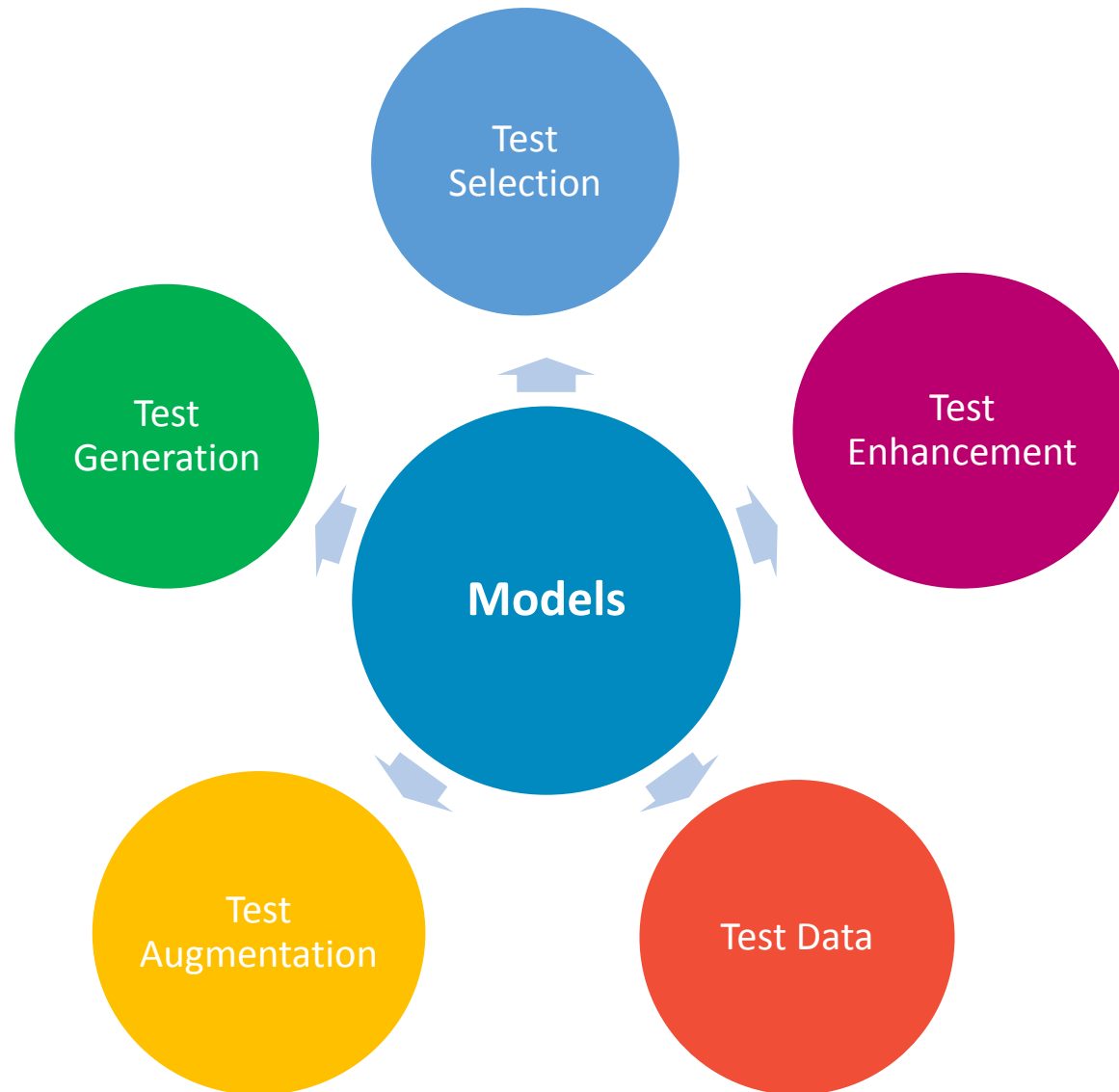
1. Point of variability
2. Interaction coverage
3. Rules \ restriction



Modeling uses advanced techniques, such as CTD to radically optimize required test cases and deliver business value by:

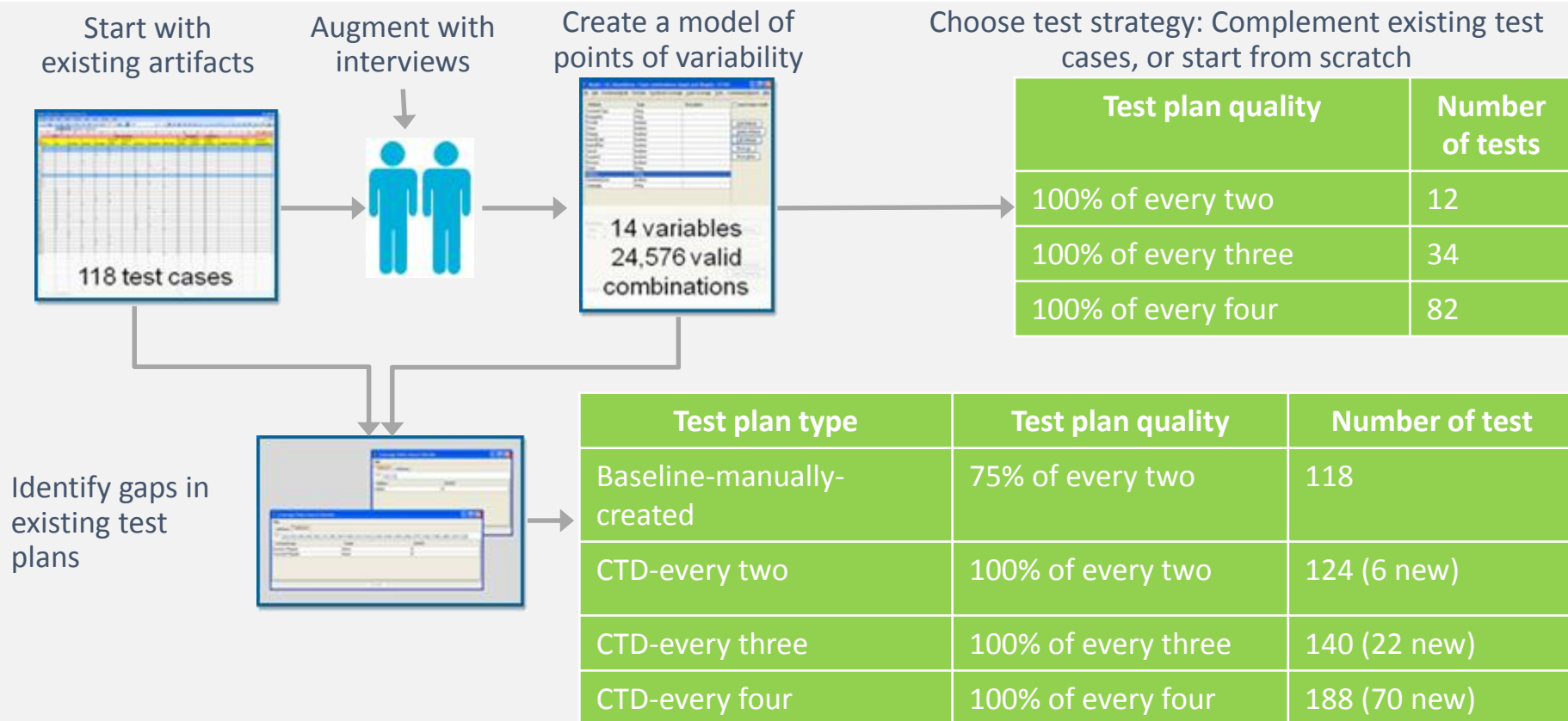


- 30% reduction in projected test case development and execution costs
- 30% reduction in schedule
- 25% improvement in test coverage



Common flow for modeling

The following diagram depicts the common flow of modeling for CTD:

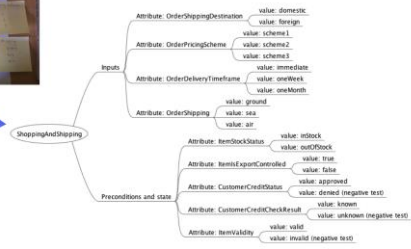


Steps for modeling using IBM's FoCuS tool

We document the test requirements and points of variation: we call them test cases.

1

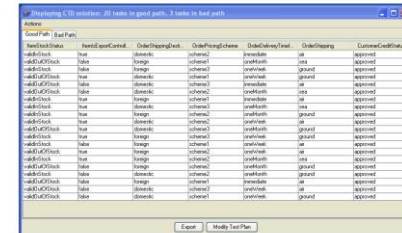
Create model



© 2011 IBM Corporation

For this new set of requirements – pairwise for the entire model, with full coverage of the shipping and export control attributes in the model – we get a test plan of only 20 test cases.

IBM



4

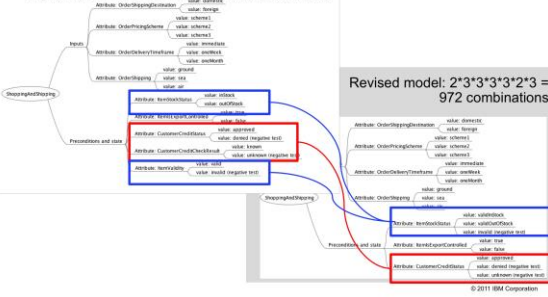
Generate Output

2

Refine model

Of the attributes and values, some values may be redundant or overlap. Discussion with subject matter experts is critical.

Original model: $2^3 \times 3^3 \times 3^2 \times 2^2 \times 2^2 = 1,728$ combinations



Revised model: $2^3 \times 3^3 \times 3^3 \times 2^3 = 972$ combinations

© 2011 IBM Corporation

And what if interaction level two isn't enough? Let's say that we have greater business or technical risk in one area of the model – how does IBM's FoCuS/CTD let us test it more intensively?

Let's tell FoCuS/CTD that we want full coverage of the interaction of the values for these four points of variation: shipping, delivery, export control and destination country, e.g.

"What if there's a bug in ground shipping with a delivery timeframe of one week for export-controlled items going to a foreign destination?"



Now FoCuS/CTD must optimize for two sets of coverage requirements:

- Interaction level two (which creates 150 detailed test coverage requirements)
- Interaction level four (which creates 20 detailed test coverage requirements)



3

Set intensity

© 2012 IBM Corporation

A c t i v i t y

Category	Description	Duration
Purpose	Capture requirements on a sheet along with questions for participants to answer.	
Participation	In table groups	2 minutes
Process	Read the requirements given above and answer the following questions: ▪ How many test cases do you estimate to create for this requirement? ▪ What scenario you think you should cover for complete testing for this requirement?	5 minutes
Product	Present to the class the input and the output	5 minutes



Activity Solution: Online shopping system

- 1 The system shall take orders for any valid item, whether it is in stock or not.
- 2 The system shall support multiple pricing schemes for an order.
 - o The first scheme ... [description omitted for demo]
 - o The second scheme ... [description omitted for demo]
 - o The third scheme ... [description omitted for demo]
- 3 The system shall validate the current credit status of the purchaser, when known.

The purchaser can select one of the following time frames for order delivery: immediate, within one working week, and within one month. Ground shipping is default, while air shipping is allowed for orders being delivered in a week or a month, and air shipping is allowed for immediate or one-week orders.
- 4 When an item is classified as export controlled, system shall generate the appropriate work items to comply with governmental requirements.

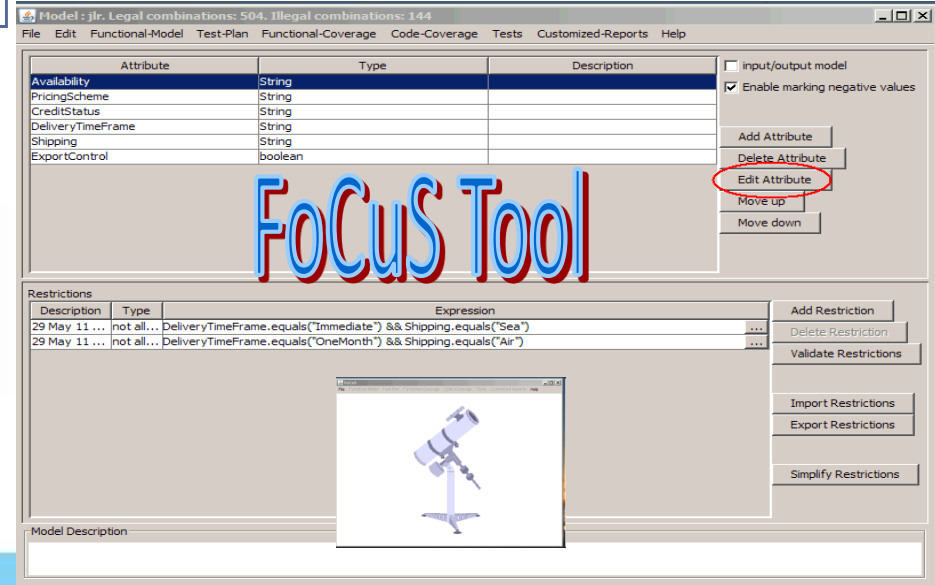
Inventory 1	Pricing 2	Credit stat 3	Delivery Time 5	Shipping 6	Customer 7
Available	Internet Banking	Good	Immediate	Ground	New
Not in Stock	Credit Card	Bad	One week	Air	Existing
Discontinued	Debit Card	unknown	One month	Sea	
No Such Product	Cash	NA	NA		

$2*4*4*5*3*3 = 1440$ combinations To Be Tested !!!

What is need for testing?
Does everything require combinatorial?
Do we really need to test all combinations?

The root cause analysis of many bugs shows they depend on a value of one variable (20% - 68%)

Most defects can be discovered in tests of the interactions between the values of two variables (65 - 97%)



CTD gave 22 Tests which were covering all interactions and with NO redundancy



Questions?

02

Which type of CTD model can create or identify a lean subset of test to cover a similar functionality?

A

Test Generation

B

Test Selection

C

Test Enhancement

D

Test Augmentation

03 Overview of Document Analytics

What is Document Analytics?

Document Analytics
is a technology
developed by Tokyo
Research Lab (TRL)

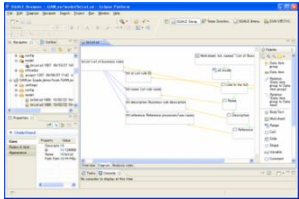
Enables various kinds of **artifact analysis** and
provides **assets** focused on **document analysis**
processes

It works especially
well when

A large amount of project documents are written
with **Microsoft Office** and written in a **specific**
format or in **combination of tables** and **natural**
sentences

Basic concept of Document Analytics

Document
Analytics
Designer User



Document Analytics Designer is a tool to define the processing and analyzing rules to deal with diversity in documents.

Extracting the theoretical structures and contents from the format specific to individual projects

Combination of basic analytical operations can create various analytical rules

Documents
to be analyzed



1

Document
modeling

2

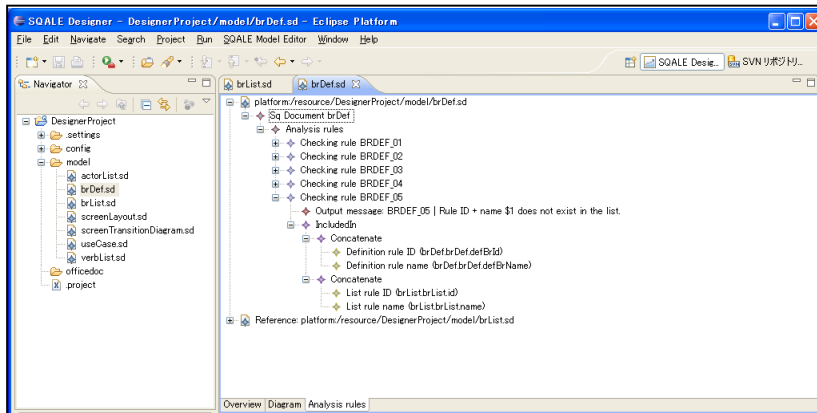
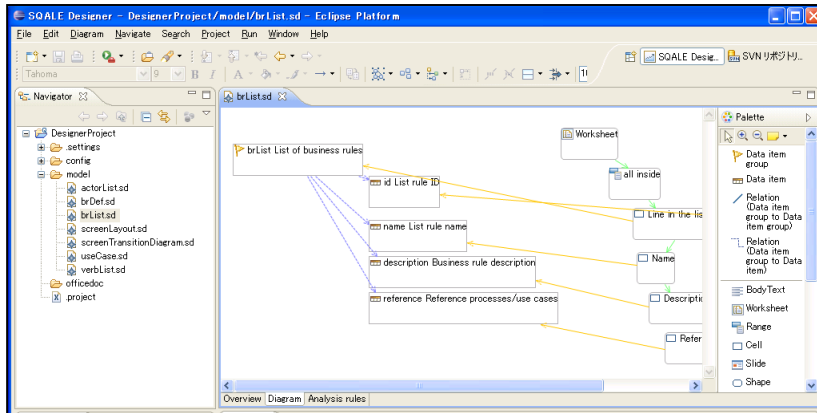
Document
models

Applying
analytical rules

Runtime: Program to perform the document analysis

Output

Compiling and execution



.jar

Runtime analysis engine

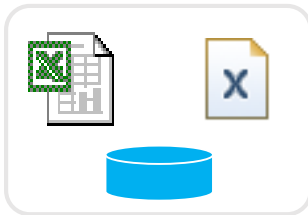
Output report



Analysis Outputs

Summary reports

Detailed error reports



Project A	Order management	macro design	business rule definition	IBM AA	2009
Rule ID	BR-OM-0006	Rule name			
Rule definition					
1. Types of unit price System defines the following 2 types of unit price: Sales unit price: unit price in the base currency unit Local unit price: unit price in the local currency 2. Unit price entered by user Unit price can be entered by the user. When the user enters the unit price, a) For local order, the entered unit price is treated as sales unit price and local unit price, without conversion b) For export order, the entered unit price is treated as local unit price and sales unit price is calculated by multiplying currency exchange rate to the local unit price 3. Rules for calculating sales unit price and local unit price					

Source C:\SQALE\sqale-work\input\1.Business Requirements\BusinessRules.xls						
User na subhmukh						
Date/tir 2012/06/26 10:59:21						
Rules e CHECK1 CHECK2 CHECK3 CHECK4 CHECK5						
Tool ver 2.2.0-20120626105910						
Error	Location Path	Seve	Tag	Rule ID	Message	Message
	201206:/Workbook/C:\SQALE\sqale-work\input\1.Business Requirements\BusinessRules.xls	Error		CHECK3	Business rule ID does not exist in the list of business rules	BR-OM-0000
	201206:/Workbook/C:\SQALE\sqale-work\input\1.Business Requirements\BusinessRules.xls	Error		CHECK5	Business rule ID+NAME does not exist in the list of business rules	BR-OM-0000,Constraint on quote and order relationships
	201206:/Workbook/C:\SQALE\sqale-work\input\1.Business Requirements\BusinessRules.xls	Error		CHECK2	Business rule Name must not be blank	
	201206:/Workbook/C:\SQALE\sqale-work\input\1.Business Requirements\BusinessRules.xls	Error		CHECK5	Business rule ID+NAME does not exist in the list of business rules	BR-OM-0006,
	201206:/Workbook/C:\SQALE\sqale-work\input\1.Business Requirements\BusinessRules.xls	Error		CHECK5	Business rule ID+NAME does not exist in the list of business rules	BR-OM-0007,Calculation of unit price



Early defect detection

Consolidate information for human review



Traceability analysis

Check for consistency across models and documents



Monitor project health



Questions?

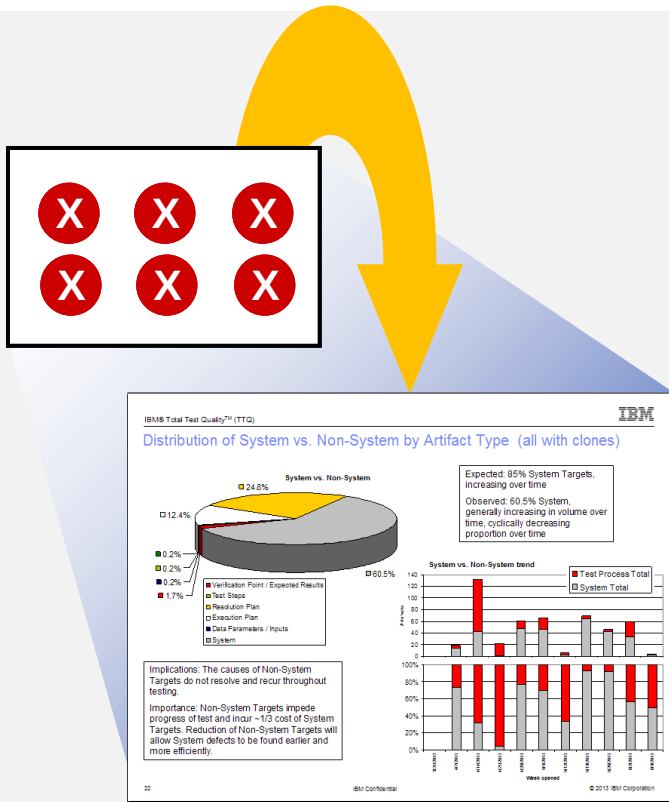
04 Overview of CDM / DAS

What is IBM Collaborative Defect Management (CDM)?



The IBM CDM uses IBM's Defect Reduction Method™ (DRM) to identify high-priority cost reduction opportunities and risks in development projects.

CDM: Recover intelligence from defect data



Key benefits:

- Detailed **assessment** of **risk** and **quality** for technical team consumption
- **High-level summaries** for executive consumption
- **Specific recommendations** to address the highest priority areas of improvement

Root Cause Analysis (RCA) versus CDM (DRM / DAS)



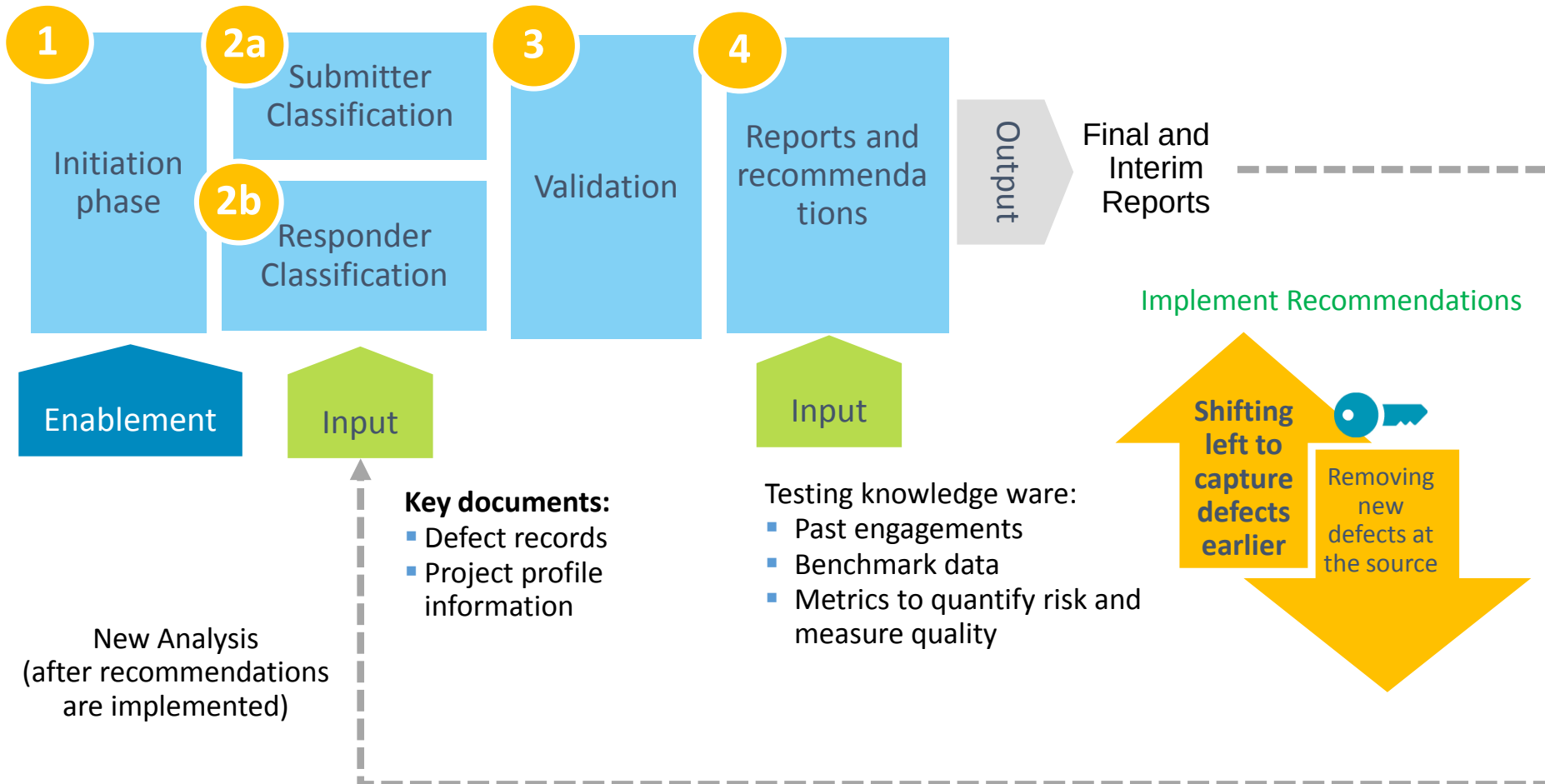
RCA

- Discovers the **root cause** of **individual defects**
- Looks at **undesirable characteristics** only
- Is **costly** and **time-consuming** and considers code-related defects only

- Looks at **larger patterns** of injection
- Discovers **good patterns** and **trends**
- Is rapid and does not require deep knowledge of the project
- Analyzes all types of **defects**, such as code, data, database, environment, local language support, process, and user documentation

CDM (DRM / DAS)

Defect Analytics as an embedded methodology



Defect Analysis categories

System

Non-System

**Not
classifiable
(test)**

**Not
classifiable
(both)**

Excluded

Milestones or Goals

Startup Phase:

- Define CDM Scope
- Perform Baseline DAS
- Define Tooling Requirements
- Implement Tool Changes
- Create Training Materials
- Provide Classification Training
- Implement Data Collection
- Establish Metrics

Perform Phase (Iterative):

- Create Assessment schedule
- Perform Defect classification
- Perform Classification validation
- Provide Validation Training
- Create Interim Reports
- Review Interim Reports

Close Phase (Iterative):

- Create Final Reports
- Review Final Reports
- Provide Metrics Collection Guidance
- Implement Improvements
- Track Metrics



Questions?

01

Which one of the following **Defect analysis categories** refers to a change that was made to some part of the system, such as environment, data, and code?

A

Non-system

B

System

C

Excluded

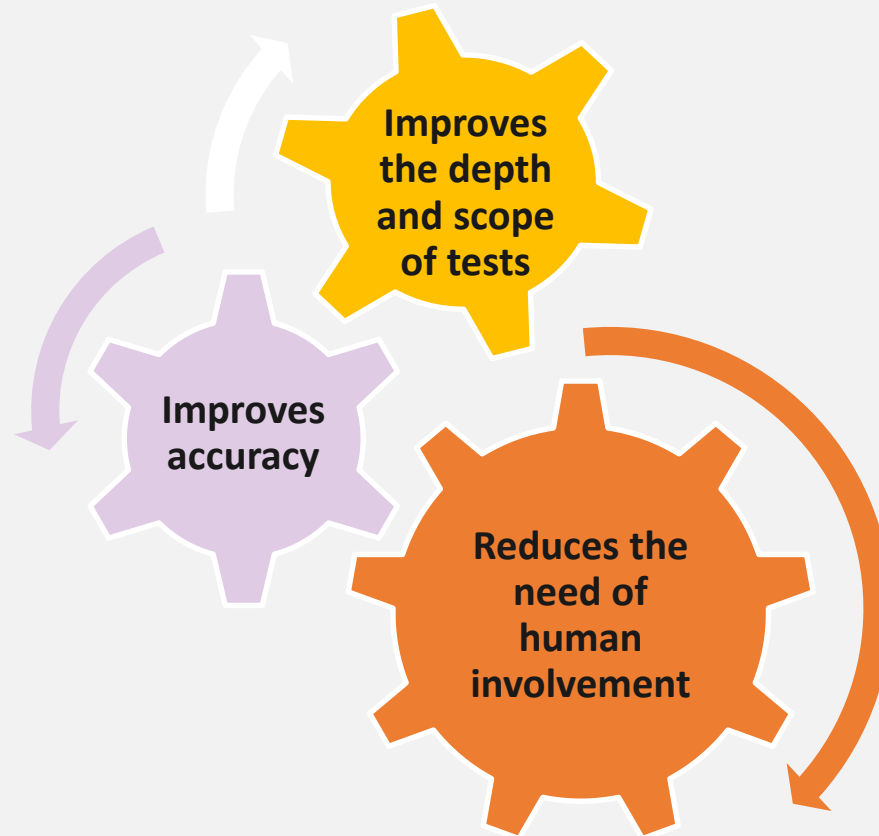
D

Not classifiable (test)

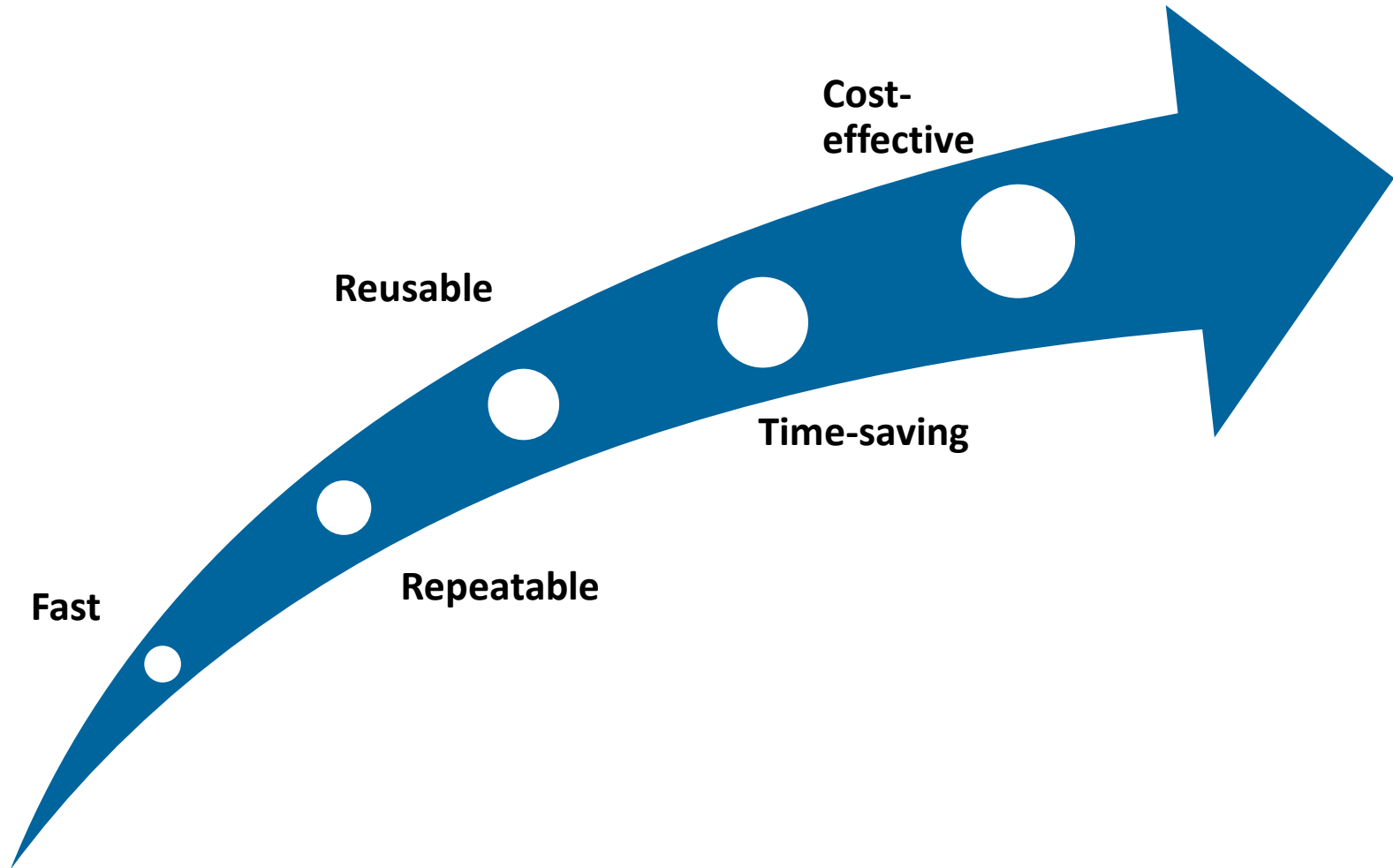
05

Overview of Adaptive Test Automation (ATA)

What is test automation?



Automation testing helps automate the manual tests using a tool or a programming language.

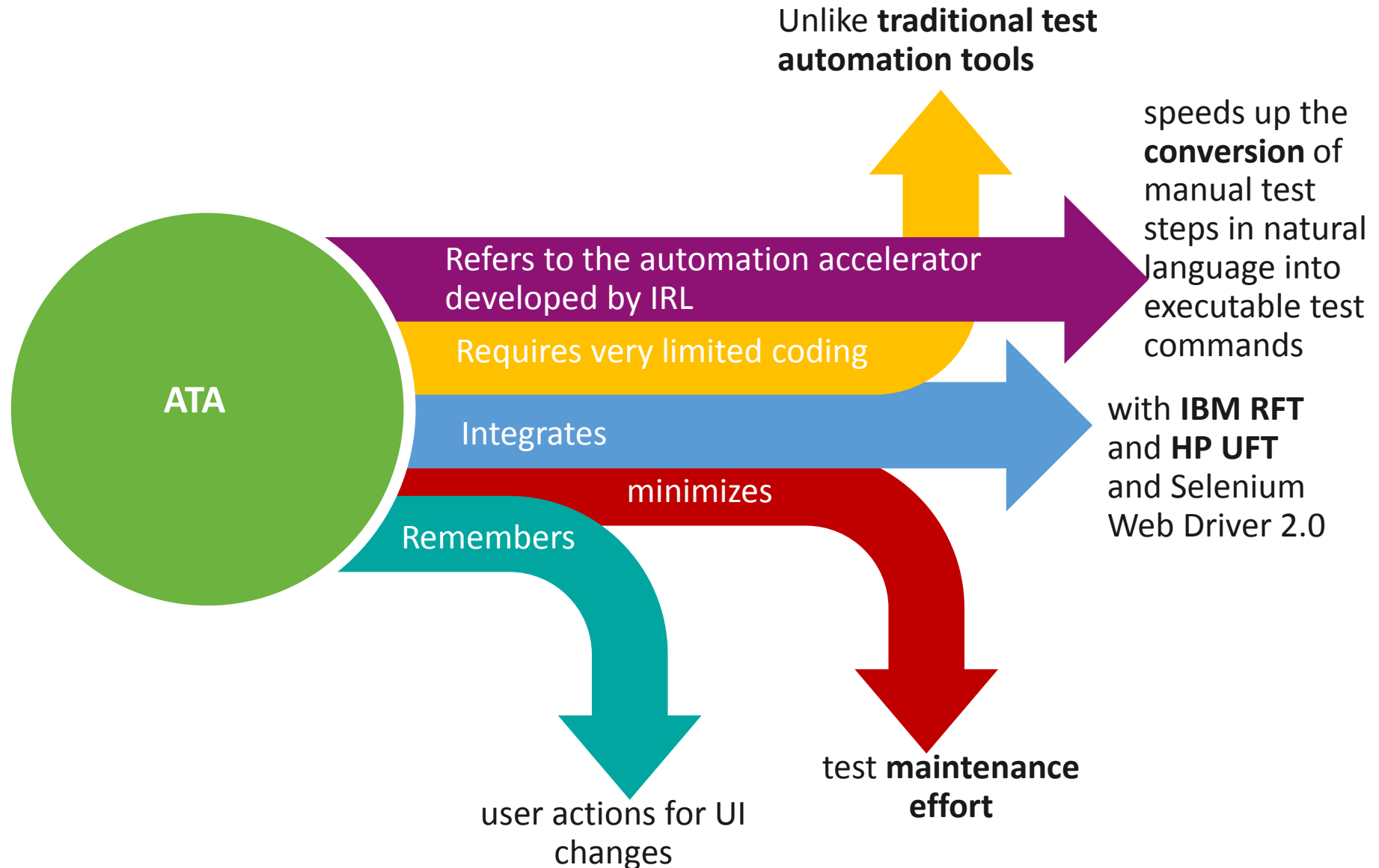


Licensing cost



Lack of skills

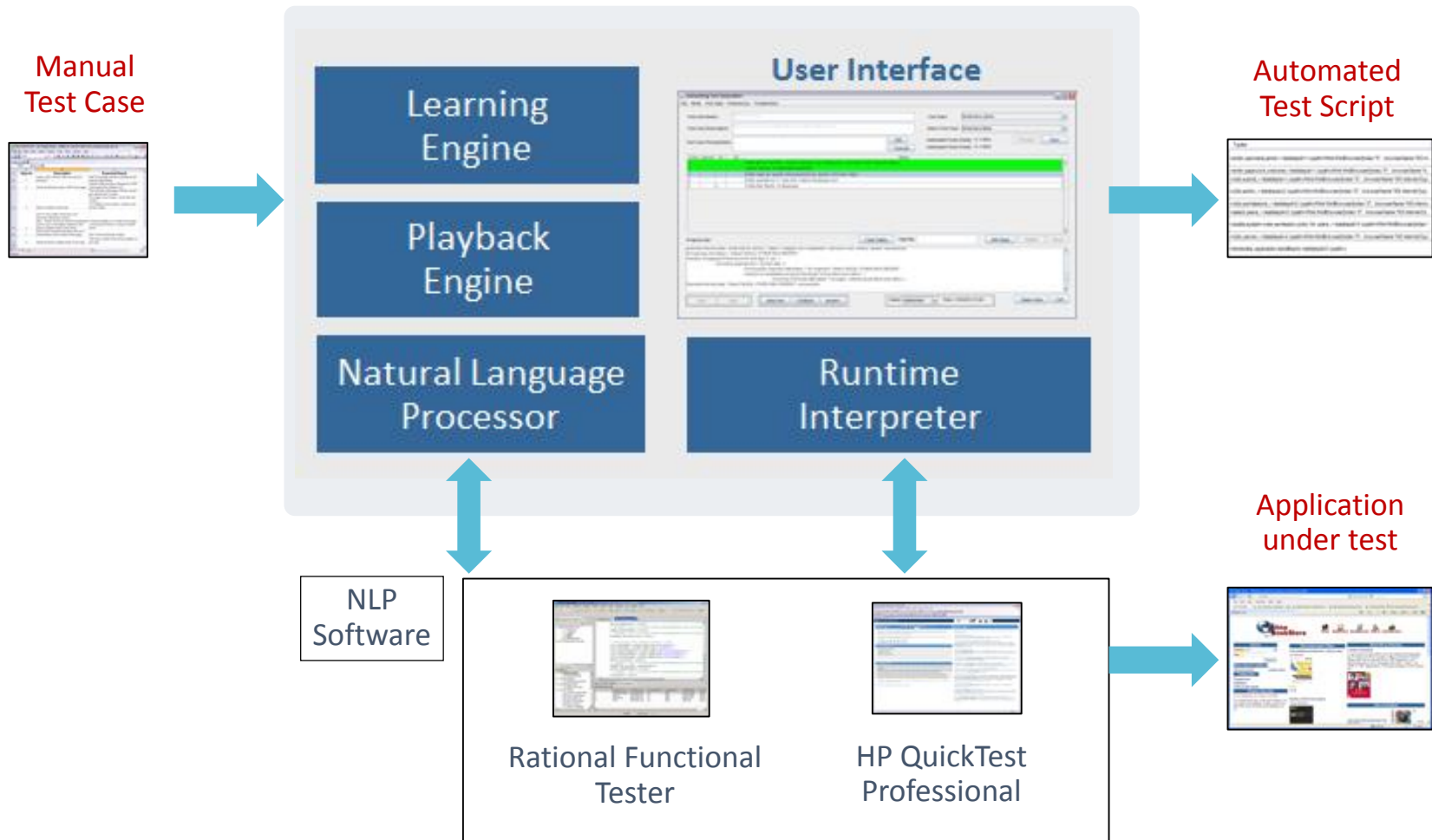
What is Adaptive Test Automation (ATA)



ATA is not a
standalone test
automation tool

ATA is a test automation accelerator that needs to be installed on top of **IBM RFT** or **HP UFT**, and Selenium Web Driver 2.0.

ATA high-level architecture



ATA Graphical User Interface (GUI)



The screenshot displays the ATA GUI with the following components and labels:

- Navigation and Suite Selection:** Test Suite: Bookstore-Demo, Select Test Case: 1_Bookstore-Main, Automated Tests (Suite): 1 / 3 (33%), Automated Tests (Total): 1 / 3 (33%), Previous, Next.
- Test Case Information:** Test Case Name: 1_Bookstore-Main, Test Case Description: Searches for book and adds the book to the shopping cart, Test Case Prerequisites: (empty).
- Manual Test Steps Table:**

S.No.	Ignore	VS	CE	Steps
1.				Enter title as "MySQL", select "Category" as "Databases", and then Click "Search" Button
2.				Select "MySQL (OTHER NEW RIDERS)"
3.				Enter "Title of name" and password, and click "login"
4.				Enter quantity as "2" and click "Add to Shopping Cart"
5.		X		Verify that "Items" exists
6.				Click "Testing" button
- Automation Controls:** Ignore step indicator, Conditional execution, Reuse tuples from existing automated scripts, Copy Tuples, Data file: (empty), Edit Steps, Update, Reset.
- Automation Log:** Executed the test step: "Enter title as mysql, select Category as Databases, and then Click Search Button" successfully. Processing Test Step 2: "Select MySQL (OTHER NEW RIDERS)". Number of segment lists found for test step 2, are: 1. Choosing segment list: 1 for test step: 2. Choosing the segment alternative: 1 for segment: "Select MySQL (OTHER NEW RIDERS)". Number of candidates found for the target "mysql (other new riders)": 1. Choosing ATDTuple alternative: 1 for tuple: <select,mysql (other new riders),,>. Executed the test step: "Select MySQL (OTHER NEW RIDERS)" successfully.
- Automation Status and View:** Automate Step, Stop, Status: UNKNOWN, Time: 09/05/2013 13:52:24, Playback View, Exit.
- Automation Mode:** ☐ Automatically proceed to the next step during automation.

ATA is applicable to web applications only

Includes traditional and AJAX applications excluding custom web components such as DOJO widgets and flash.

Manual test case format

- Translate to the Excel format expected by the default importer
- Rewrite manual tests directly in ATA
- Implement a custom importer

Import existing automated test scripts

- ATA currently does not have the capability to import automated test scripts created in QTP/RFT
- Such scripts would have to either be re-automated in ATA or be outside the scope of automation with ATA

Leverage existing automation frameworks

- Clients often have existing automation frameworks. Reusable functions in such frameworks can be leveraged through ATA's capability to call functions written in QTP / RFT.
- However, the functions might need some refactoring for Naming convention, Types / number of formal parameters, and References to object repository (QTP / RFT).

If an automation framework exists, ATA may not provide significant benefits

- ATA [Wiki](#)
- ATA [Self-paced training](#)
- ATA [Community](#)
- ATA SAL Contact: Saritha Route

